Detaily charakterizace semiempirického modelu a srovnání identifikovaných modelů MOSFET EKV 2.6 a BSIM 3.3 (ON Semiconductor)

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Implementace analogových soustav

1 Detaily jednokrokové extrakční procedury pro semiempirický model

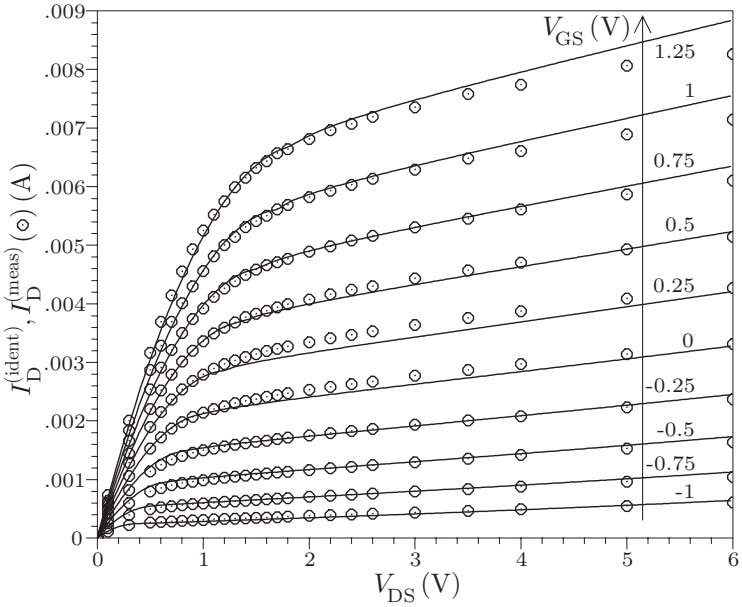
1.1 Tranzistory s indukovaným kanálem

Parameter		2N3608				BUZ345			
Symbol	Unit	Estim.	Min.	Max.	Final	Estim.	Min.	Max.	Final
V_{TO}	V	−4	−8	−2	−4.77	3.3	3.2	3.35	3.26
ϕ_S	V	0.6	0.2	1.1	0.657	0.6	0.4	0.8	0.578
ϕ_O	V	0.8	0.3	1.2	0.806	0.8	0.7	0.9	0.801
W	µm	100	1	2000	37.9	1.4×10^6	5×10^5	1.5×10^6	1.46×10^6
L	µm	5	1.5	50	3.46	2	1	500	4.97
X_J	µm	1	0.2	10	1.54	0.1	0.002	5	0.289
X_{JL}	µm	0.8	0.1	8	0.762	0.01	0.0002	5	0.179
t_{ox}	nm	100	10	200	98.7	100	10	500	74.7
N_{FS}	m^{-2}	10^{15}	10^{11}	10^{18}	10^{15}	10^{15}	10^{13}	10^{18}	10^{15}
N_A	m^{-3}	10^{20}	10^{18}	10^{23}	2.32×10^{22}	10^{21}	10^{19}	10^{24}	1.73×10^{20}
$v_{\max}$	m/s	5×10^4	5×10^3	10^6	3.55×10^5	3×10^4	10^3	10^6	3.23×10^5
μ_O	$\text{m}^2/(\text{Vs})$	0.08	0.03	0.12	0.0719	0.06	0.04	0.08	0.0585
E_P	MV/m	4	2	6	3.4	0.0001	0*	0.1	0
$\varkappa$	–	0.2	0.02	0.9	0.441	0.03	0.001	0.1	0.0306
K_P	A/V^2	3×10^{-5}	5×10^{-6}	2×10^{-4}	2.49×10^{-5}	4×10^{-5}	2×10^{-5}	5×10^{-5}	4.19×10^{-5}
γ	$\sqrt{\text{V}}$	0.4	0.15	0.8	0.294	0.3	0.0001	0.6	0.366
δ	–	1	0.4	2.5	0.989	1	0	1.2	1
η	–	1	0.03*	30	0.03	0.1	0*	1.2	0
θ	V^{-1}	0.05	0.001	0.2	0.00334	0.05	0	0.1	0.0384
ι	–	1	0.1	10	0.34	1	0	3	0.572
r_D	Ω	–	–	–	10	0.03	0.01	0.05	0.0249
r_S	Ω	–	–	–	10	0.07	0.02	0.1	0.0435

1.2 Tranzistory se zabudovaným kanálem

Parameter		KF521 direct current & capacitance model			
Symbol	Unit	Estim.	Min.	Max.	Final
V_{TO}	V	−1.25	−2	−1	−1.48
ϕ_S	V	0.6	0.3	1	0.334
ϕ_O	V	0.8	0.4	1	0.789
W	µm	550	300	900	443
L	µm	5	2	10	4.83
X_J	µm	1	0.5	2	0.932
X_{JL}	µm	0.8	0.2	1.5	0.827
t_{ox}	nm	100	20	200	71.8
N_{FS}	m^{-2}	10^{15}	10^{13}	10^{17}	10^{15}
N_A	m^{-3}	10^{21}	10^{19}	10^{23}	7.51×10^{21}
v_{max}	m/s	5×10^4	10^4	5×10^5	1.71×10^5
μ_O	$\text{m}^2/(\text{Vs})$	0.08	0.02	0.2	0.0535
E_P	MV/m	0.8	0.3	1	0.419
$\varkappa$	–	0.2	0.05	0.4*	0.4
K_P	A/V^2	3×10^{-5}	10^{-5}	5×10^{-5}	2.12×10^{-5}
γ	$\sqrt{\text{V}}$	0.4	0.2	0.6	0.568
δ	–	1	0.6	1.3	1
η	–	1	0.5	2	0.811
θ	V^{-1}	0.05	0.002*	0.2	0.002
ι	–	1	0.5	1.5	0.929
r_D	Ω	10	1	20	11.8
r_S	Ω	10	1	20	5.17
$C_{JO} \times \text{area}_S$	pF	2.2	1.2	2.8	2.23
$C_{JO\text{sw}} \times \text{perimeter}_S$	pF	0.2	0.1	1	0.275
$C_{JO} \times \text{area}_D$	pF	1.6	1.3	1.8	1.58
$C_{JO\text{sw}} \times \text{perimeter}_D$	pF	0.2	0.01	0.5	0.188
ϕ_{OS}	V	0.7	0.55	1	0.556
ϕ_{OD}	V	0.7	0.6	0.8	0.695
m_S	–	0.5	0.1	0.6	0.265
$m_{S\text{sw}}$	–	0.3	0.2	0.5	0.283
m_D	–	0.15	0.1	0.5	0.205
$m_{D\text{sw}}$	–	0.285	0.2	0.35	0.286

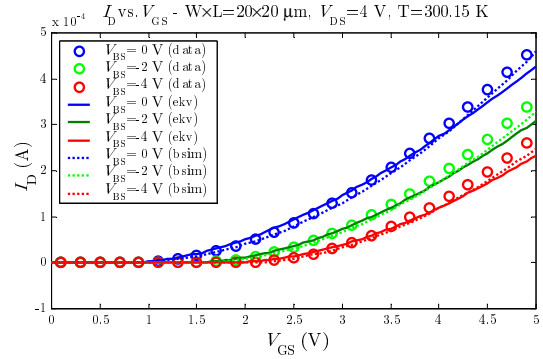
1.2.1 KF521 – charakteristický výsledek extrakční procedury (semiempirický model)

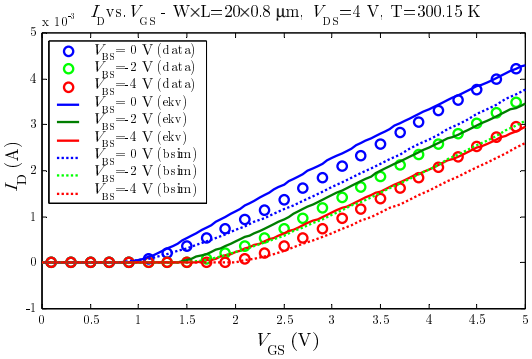


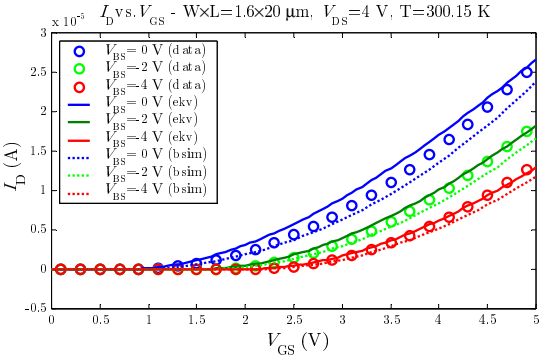
Celkový počet . . .	2N3608	BUZ345	KF521		
			DC	c_{BS}	c_{BD}
akceptovaných iterací	43	39	155	12	8
zamítnutých iterací	23	21	78	7	5
logaritmických tlumení	33	25	149	5	0

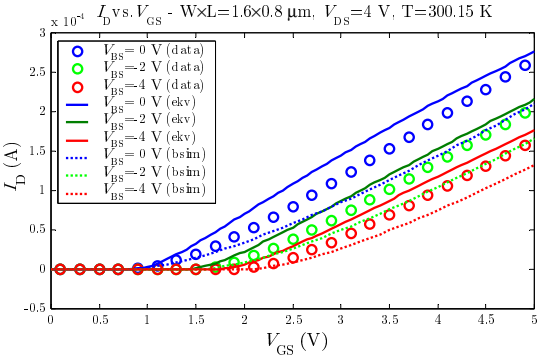
2 EVK vs. BSIM – srovnání pro čtyři zásadně různě dimenzované prvky

2.1 I_D versus V_{GS}

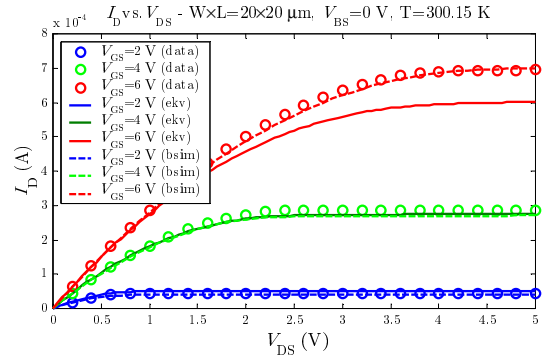


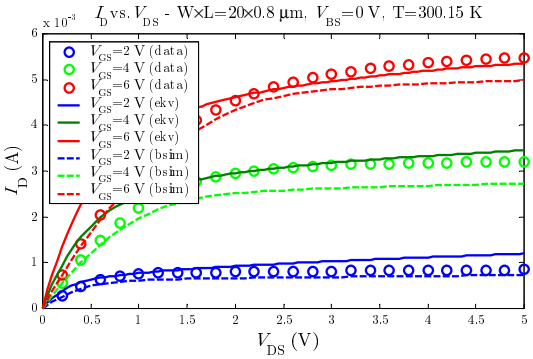


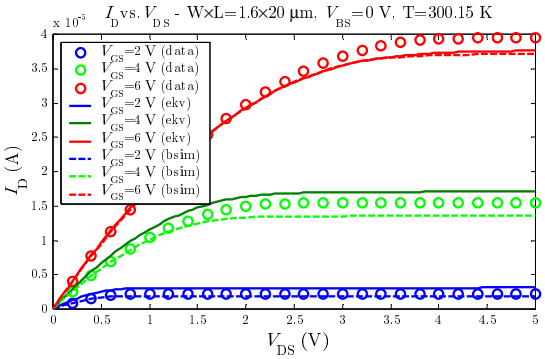


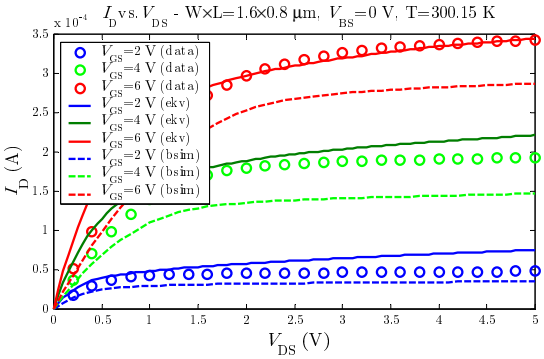


2.2 I_D versus V_{DS}

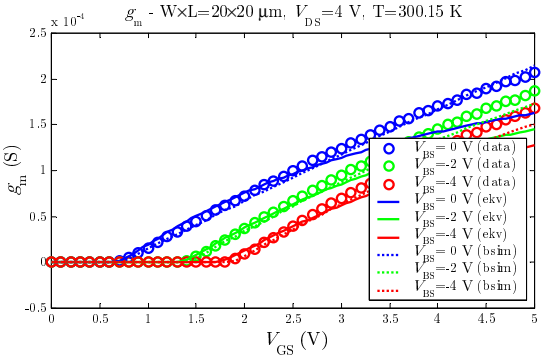


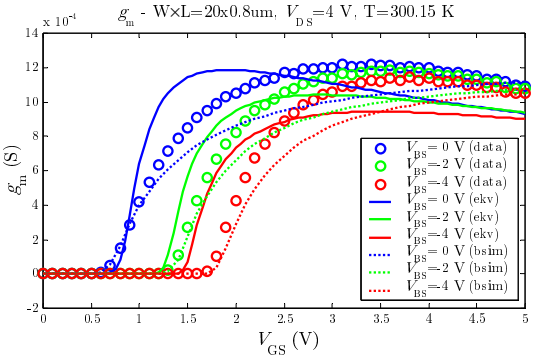


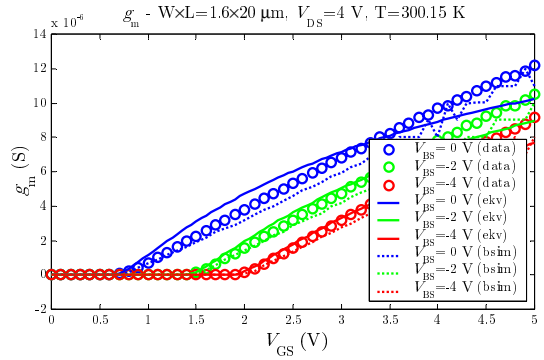


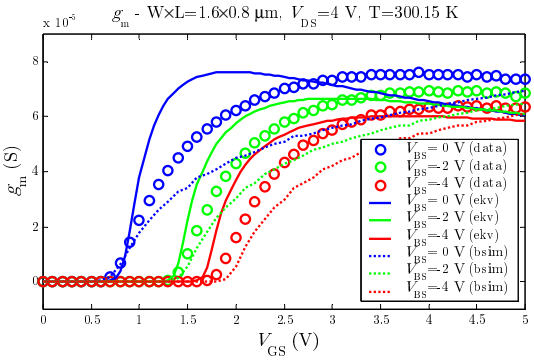


2.3 g_m versus V_{GS}

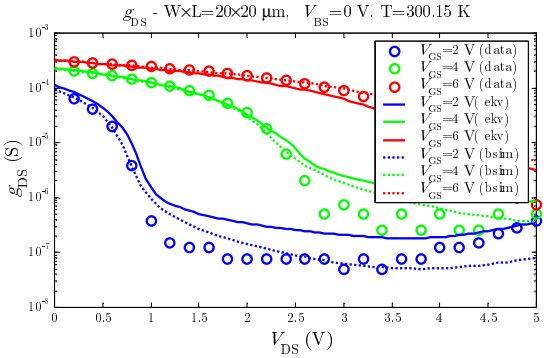


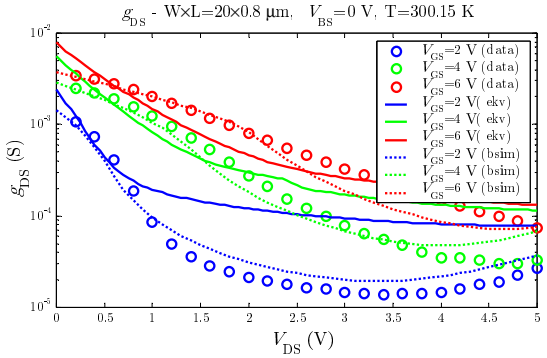


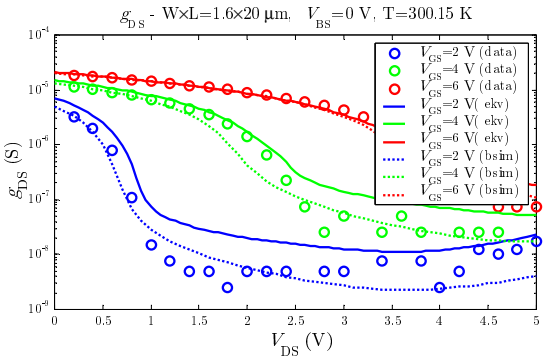


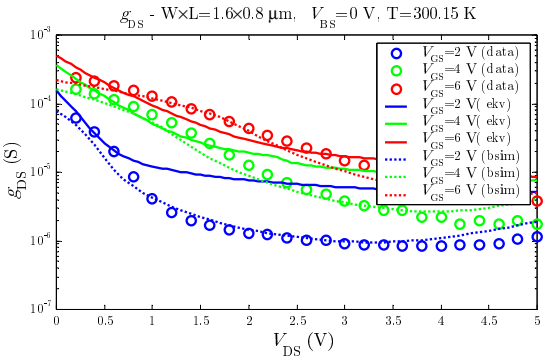


2.4 g_{DS} versus V_{DS}









3 Detaily vícekrokové extrakční procedury pro modely EKV a BSIM

3.1 EKV 2.6

Parameter		N-channel MOSFET of the ON Semiconductor BiCMOS technology (field oxide 20 nm LOCOS)					
Symbol	Unit	Estim.	Min.	Max.	Final	Device dimensions	Characteristics
V_{TO}	V	0.5	0	2	0.755	wide long	V_P vs. V_{GS}
γ	$\sqrt{\mathrm{V}}$	1	0	2	0.7	wide long	V_P vs. V_{GS}
ϕ	V	0.7	0.3	2	0.4103	wide long	V_P vs. V_{GS}
K_P	$\mathrm{A/V^2}$	5×10^{-5}	10^{-5}	∞	1.5687×10^{-4}	wide long	I_{D} vs. V_{GS}
E_O	V/m	10^{12}	10^{5}	∞	9.0881×10^{6}	wide long	I_{D} vs. V_{GS}
I_{BA}	$\mathrm{m^{-1}}$	0	0	5×10^{8}	6.4486×10^{7}	wide long	I_{B} vs. V_{GS}
I_{BB}	V/m	3×10^{8}	1.8×10^{8}	4×10^{8}	1.8×10^{8}	wide long	I_{B} vs. V_{GS}
I_{BN}	–	1	0.4	10	1.1	wide long	I_{B} vs. V_{GS}
L_η	–	0.1	0	2	0.50294	wide short and long	V_P vs. V_{GS}
Q_O	$\mathrm{As/m^2}$	0	0	∞	3.3998×10^{-4}	wide short and long	V_P vs. V_{GS}
L_K	µm	0.29	0.05	2	0.55398	wide short and long	V_P vs. V_{GS}
U_{crit}	V/m	2×10^{6}	10^{5}	2.5×10^{7}	1.1777×10^{6}	wide short	I_{D} vs. V_{DS}
λ	–	0.5	0	3	1.1311	wide short	I_{D} vs. V_{DS}
W_η	–	0.25	0	2	0.35	narrow long	V_P vs. V_{GS}
ΔW	m	0	$-\frac{1}{2}W_{\mathrm{min}}$	$\frac{1}{2}W_{\mathrm{min}}$	-2.8827×10^{-7}	different widths	I_{D} vs. V_{GS} @ low V_{DS}
ΔL	m	0	$-\frac{1}{2}L_{\mathrm{min}}$	$\frac{1}{2}L_{\mathrm{min}}$	-9.9868×10^{-8}	different lengths	I_{D} vs. V_{GS} @ low V_{DS}

3.2 BSIM 3.3 (1. část)

Parameter		N-channel MOSFET of the ON Semiconductor BiCMOS technology (field oxide 20 nm LOCOS)						Extr.
Name	Unit	Estim.	Min.	Max.	Final	Device dims	Characteristics	step
vth0	V	0.7	−2	2	0.7623	large	I_D vs. V_{GS} @ low V_{DS}	1
k1	$\sqrt{\text{V}}$	0.5	−2	2	0.9418	large	I_D vs. V_{GS} @ low V_{DS}	1
k2	–	0	−0.5	0.5	−0.0143	large	I_D vs. V_{GS} @ low V_{DS}	1
u0	$\text{cm}^2/(\text{Vs})$	670	10^{-3}	2×10^{3}	540.5	large, short	I_D vs. V_{GS} @ low V_{DS}	1
ua	m/V	2.25×10^{-9}	-10^{-6}	10^{-6}	3.17×10^{-10}	diff. widths	I_D vs. V_{GS} @ low V_{DS}	1, 2
ub	$(\text{m/V})^2$	5.87×10^{-19}	-10^{-12}	10^{-12}	2.87×10^{-18}	diff. widths	I_D vs. V_{GS} @ low V_{DS}	1, 2
uc	$(\text{m/V})^2$	−0.046	−0.1	0.1	0.034	diff. widths	I_D vs. V_{GS} @ low V_{DS}	1, 2
voff	V	−0.08	−0.5	0.5	−0.1078	large	I_D vs. V_{GS} @ low V_{DS}	1
nfactor	–	1	0.1	10	0.9445	large	I_D vs. V_{GS} @ low V_{DS}	1
prwg	1/V	0	−1	20	−0.044	short	I_D vs. V_{GS} @ low V_{DS}	2
prwb	$\text{V}^{-0.5}$	0	−1	1	0.0882	short	I_D vs. V_{GS} @ low V_{DS}	2
rdsw	Ωμm	0	0	5×10^{3}	2700	short, small	I_D vs. V_{GS} @ low V_{DS}	2, 7
a0	–	1	0	10	1.36	large	I_D vs. V_{GS} @ high V_{DS}	3
ags	1/V	0	−1	1	0.164	large	I_D vs. V_{GS} @ high V_{DS}	3
keta	1/V	−0.047	−0.1	0.1	1.577×10^{-3}	large	I_D vs. V_{GS} @ high V_{DS}	3
nlx	m	1.74×10^{-7}	−1	1	3.36×10^{-8}	diff. lengths	V_{TH} vs. L_G @ low V_{DS}	4
dvt0	–	2.2	0	10	10	diff. lengths	V_{TH} vs. L_G @ low V_{DS}	4
dvt1	–	0.53	0.05	2	0.7209	diff. lengths	V_{TH} vs. L_G @ low V_{DS}	4
dvt2	1/V	−0.032	−3	1	0.0106	diff. lengths	V_{TH} vs. L_G @ low V_{DS}	4
lint	m	0	-5×10^{-7}	5×10^{-7}	-1.01×10^{-7}	diff. lengths	I_D vs. V_{GS} @ low V_{DS}	4
wint	m	0	-5×10^{-7}	5×10^{-7}	2.85×10^{-7}	diff. widths	I_D vs. V_{GS} @ low V_{DS}	5, 8
k3	–	80	−100	100	10.08	diff. widths	V_{TH} vs. W_G @ low V_{DS}	6
w0	m	2.5×10^{-6}	0	10^{-5}	3.5×10^{-6}	diff. widths	V_{TH} vs. W_G @ low V_{DS}	6
k3b	1/V	0	−10	10	−8.558	diff. widths	V_{TH} vs. W_G @ low V_{DS}	6
vsat	m/s	8×10^{4}	3×10^{4}	3×10^{5}	5×10^{4}	short	I_D vs. V_{DS}	10

3.3 BSIM 3.3 (2. část)

Parameter		N-channel MOSFET of the ON Semiconductor BiCMOS technology (field oxide 20 nm LOCOS)						Extr.
Name	Unit	Estim.	Min.	Max.	Final	Device dims	Characteristics	step
a1	1/V	0	0	1	4.44×10^{-16}	short	I_D vs. V_{DS}	10
a2	–	1	0.01	1	0.99	short	I_D vs. V_{DS}	10
pclm	–	1.3	10^{-3}	10	2.7	short	I_D vs. V_{DS}	10
pdiblc1	–	0.39	10^{-3}	2	0.4	short	I_D vs. V_{DS}	10
pdiblc2	–	0.0086	10^{-6}	1	10^{-4}	short	I_D vs. V_{DS}	10
alpha0	m/V	0	-10^{-5}	10^{-5}	4.77×10^{-6}	short	I_D vs. V_{DS}	10
alpha1	1/V	0	0	100	10^{-18}	short	I_D vs. V_{DS}	10
beta0	V	30	10^{-9}	100	37.72	short	I_D vs. V_{DS}	10
pscbe1	V/m	4.24×10^{8}	10^{7}	10^{9}	1.32×10^{8}	short	I_D vs. V_{DS}	10
pscbe2	m/V	1×10^{-5}	10^{-7}	10^{-3}	3.52×10^{-7}	short	I_D vs. V_{DS}	10
eta0	–	0.08	0	5	4.44×10^{-17}	diff. lengths	V_{TH} vs. L_G @ high V_{DS}	9
etab	1/V	−0.07	−2	2	6.989×10^{-4}	diff. lengths	V_{TH} vs. L_G @ high V_{DS}	9
dsub	–	0.2	0.1	3	0.1206	diff. lengths	V_{TH} vs. L_G @ high V_{DS}	9

Věnujte prosím pozornost poslednímu sloupci tabulky: je zjevné, že vzhledem k velké složitosti modelu

BSIM není prakticky možné všechny parametry určit najednou a proto se stanovují sérií (v tomto případě deseti) optimalizací (kroků extrakce). Výsledek předchozí extrakce vždy slouží jako startovní bod pro následující extrakci (tj. i dříve stanovené parametry se postupně dále v následujících krocích upřesňují).

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